



TECNOLÓGICO
NACIONAL DE MÉXICO



INSTITUTO TECNOLÓGICO DE VERACRUZ

Carrera:

Ing. en Sistemas Computacionales

Asignatura:

Administración de base de datos

Clave:

6J3

Grupo: A

Alumno:

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Práctica:

Creación de CRUD mediante Neo4j

Fecha de entrega: 4 de Octubre

Datos principales:

- Instalado y configurado Neo4J dentro de un contenedor Docker

Proceso:

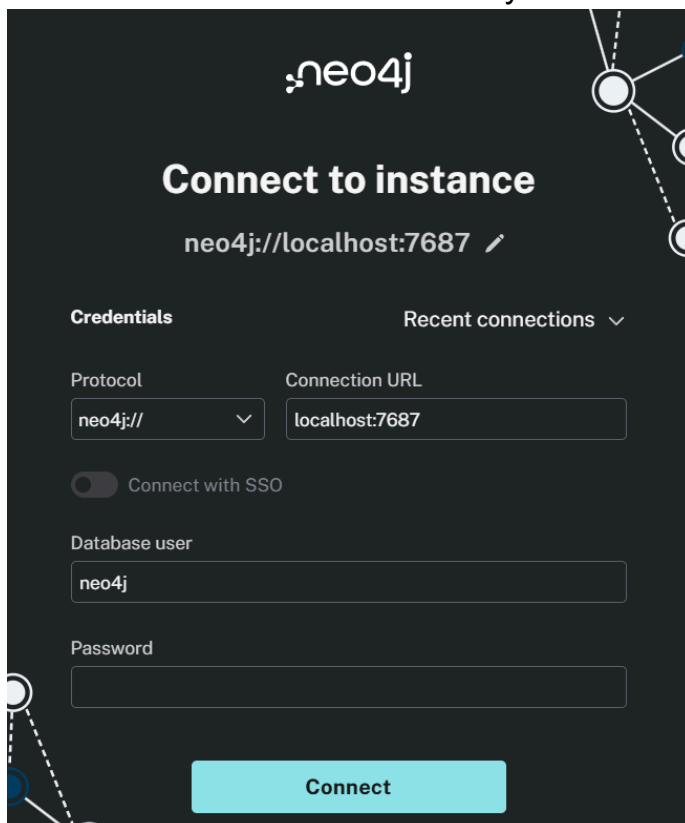
- Ejecutamos la terminal de ubuntu y encendemos el docker, posteriormente, revisamos si la imagen de neo4j esta activa

```
danett@L0V3DaNce:~$ docker start neo4j
neo4j
danett@L0V3DaNce:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
b5c4ffd8672f	neo4j:latest	"tini -g -- /startup..."	7 minutes ago	Up 7 minutes	7473-7474/tcp, 7687/tcp
a843853cea26	neo4j:latest	"tini -g -- /startup..."	8 hours ago	Up 16 seconds	0.0.0.0:7474->7474/tcp, [::]:7474->7474/tcp, 0.0.0.0:7687->7687/tcp, [::]:7687->7687/tcp

```
neo4j
```

- Una vez activado, entramos a nuestra página local(en nuestro navegador) “http://localhost:7474” .Introducimos el nombre de usuario de nuestra base de datos y contraseña



The image shows the Neo4j web interface for connecting to an instance. The title is "Connect to instance" with the URL "neo4j://localhost:7687". Below this, there are two tabs: "Credentials" (selected) and "Recent connections". Under the "Credentials" tab, there are fields for "Protocol" (set to "neo4j://"), "Connection URL" (set to "localhost:7687"), "Database user" (set to "neo4j"), and "Password" (empty). There is also a toggle for "Connect with SSO" which is currently off. A "Connect" button is at the bottom.

- Dentro de nuestra interfaz gráfica de Neo4j, existe una herramienta de consola. En este punto empezamos por **Crear 3 personas en una tabla de Alumnos**.

The screenshot shows the Neo4j console interface. At the top, there's a user dropdown set to 'neo4j' and a link to 'Connect your instance to Aura console for more tools and features'. Below this is a toolbar with 'Stream', 'CRUD', and a '+' icon. The main area contains a Cypher query:

```
1 CREATE
2   (a1:Alumno {
3     nombre: "Daniela",
4     semestre: 8,
5     carrera: "Ingenieria en Sistemas computacionales"
6   }),
7   (a2:Alumno {
8     nombre: "Axiel",
9     semestre: 9,
10    carrera: "Ingenieria Mecatronica"
11  }),
12   (a3:Alumno {
13     nombre: "Renata",
14     semestre: 5,
15     carrera: "Ingenieria industrial"
16   })
17 RETURN a1, a2, a3;
```

Below the query editor, there are tabs for 'Graph', 'Table', and 'Raw'. The 'Graph' tab is selected, showing a visual representation of the three nodes: Daniela, Axiel, and Renata. To the right, the 'Results overview' panel shows 'Nodes (3)' and a summary of the results: 3 nodes of type 'Alumno'. At the bottom, a status bar indicates 'Created 3 nodes, set 9 properties, added 3 labels' and 'Started streaming 1 record after 52 ms and completed after 73 ms.'

- Después con el comando de MATCH leemos los datos dentro de la tabla que creamos

The screenshot shows the Neo4j console interface with a new query. The toolbar now includes 'Stream', 'CRUD', and 'Untitled Query'. The Cypher query is:

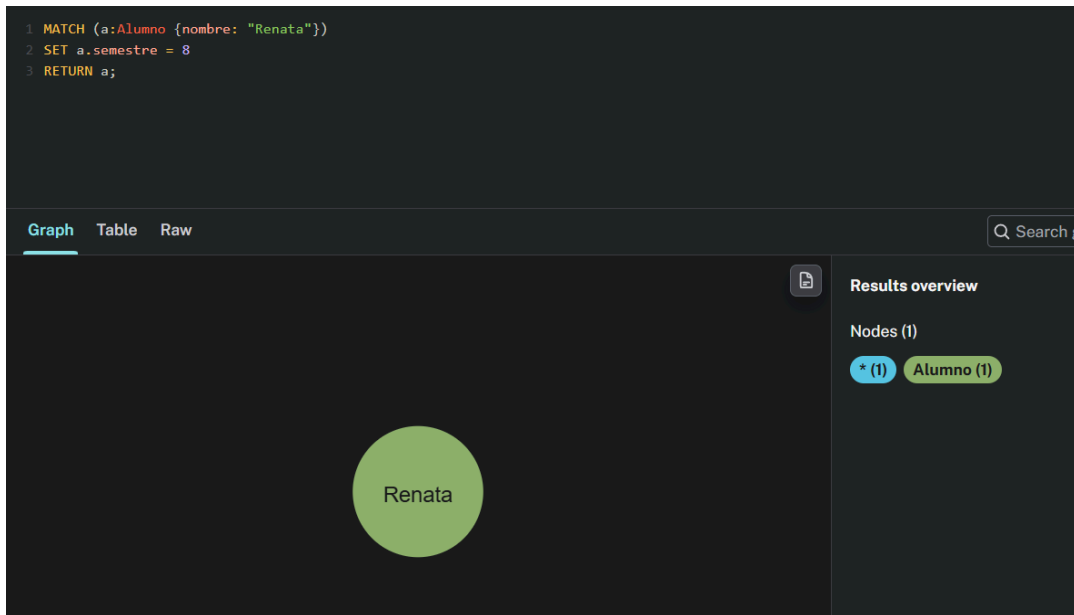
```
1 MATCH (a:Alumno)
2 RETURN a;
```

The 'Graph' tab is selected, showing the same three nodes. To the right, the 'Node details' panel is open, showing the details for the selected node 'Alumno'. It displays a table with the following data:

Key	Value
<id>	4:a76b4a95-3872-4b1f-8f3c-7708333be024:2
carrera	"Ingenieria industrial"
nombre	"Renata"
semestre	5

At the bottom, a status bar indicates 'Started streaming 3 records after 2 ms and completed after 3 ms'.

- En esta misma tabla actualizamos los datos y modificamos el semestre a la alumna “Renata”



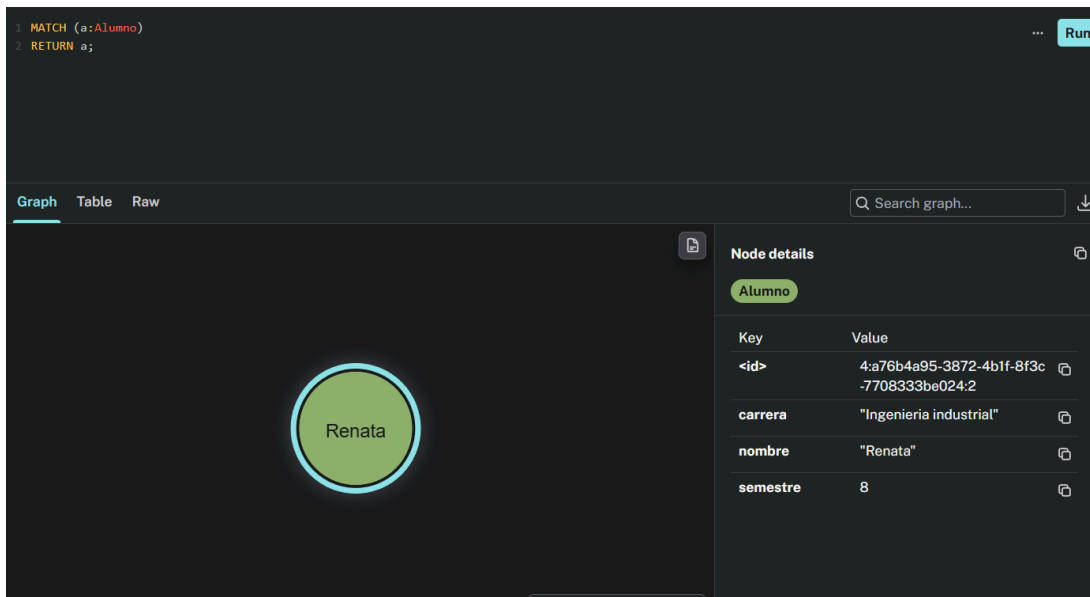
```

1 MATCH (a:Alumno {nombre: "Renata"})
2 SET a.semestre = 8
3 RETURN a;

```

The interface shows a query editor with the above Cypher query. Below the editor, there are tabs for 'Graph', 'Table', and 'Raw'. The 'Graph' tab is selected, displaying a single green node labeled 'Renata'. On the right side, there is a 'Results overview' panel showing 'Nodes (1)' with a count of 1 for the 'Alumno' type.

- Verificamos la actualización de los datos, volviendo a leer los datos



```

1 MATCH (a:Alumno)
2 RETURN a;

```

The interface shows a query editor with the above Cypher query. Below the editor, there are tabs for 'Graph', 'Table', and 'Raw'. The 'Graph' tab is selected, displaying a single green node labeled 'Renata'. On the right side, there is a 'Node details' panel showing the details for the 'Alumno' node. The details include a table with the following data:

Key	Value
<id>	4:a76b4a95-3872-4b1f-8f3c-7708333be024:2
carrera	"Ingeniería industrial"
nombre	"Renata"
semestre	8

- Finalizamos removiendo datos de la tabla creada. Leemos la tabla, sucesivamente la columna y después el nombre específico que esté en esa columna para después eliminar ese dato de la tabla .

1 MATCH (a:Alumno {nombre: "Daniela"})

2 DELETE a;

3

4 //MATCH (a:Alumno {nombre: "Axiel"})

5 DELETE a;

6

7 MATCH (a:Alumno)

8 RETURN a;

...

Run

GraphTableRaw

Q Search graph...⌵

⌵

Results overview

Nodes (1)⌵

* (1)

Alumno (1)

Renata

⌵⌵⌵⌵⌵⌵

Started streaming 1 record after 64 ms and completed after 66 ms.